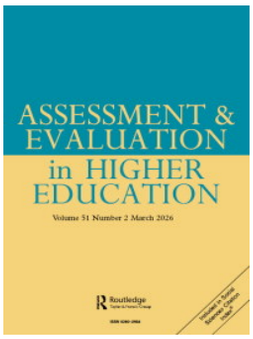

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Using AI to generate formative feedback in doctoral education

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ABSTRACT
This paper examines how artificial intelligence (AI) tools enhance feedback practices in doctoral education by providing a supplementary source of formative assessment. It explores the use of Generative AI alongside Grainger's Formative Assessment Criteria-Based Tool (F.A.C.T.) in a single-subject case study of a pre-confirmation doctoral candidate at an Australian university. The study employs a naturalistic and interpretive approach with a sequential design, exploring interactions between a doctoral student and ChatGPT across multiple sessions where the AI tool evaluated a pre-confirmation thesis. Data collection included deidentified summarised feedback received from AI and an independent academic reviewer. Findings revealed that AI-generated feedback, guided by Grainger's F.A.C.T. demonstrated thematic alignment with a human reviewer in identifying areas needing improvement, particularly regarding theoretical foundation and contribution to knowledge. However, the human reviewer provided contextually nuanced and discipline-specific feedback with more actionable suggestions. The study illustrates how, when coupled with formative rubrics, generative AI may serve as a supplementary feedback mechanism that may reduce power imbalances inherent in supervisor/reviewer-student relationships while providing expedient formative feedback. This research contributes to understanding how reflective practice in doctoral education may be enhanced through AI integration, addressing gaps in feedback literacy, socialisation, and standardised assessment parameters in doctoral contexts.

KEYWORDS
Assessment; doctoral education; feedback literacy; artificial intelligence

Introduction

Generative artificial intelligence (AI) tools are being used by students in various ways in the tertiary education environment, particularly in undergraduate contexts but less so in the doctoral environment. Students use such tools to generate ideas,

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source literature, analyse data, refine writing, summarise texts and create reference lists (Parkinson and Reinke 2024). In doctoral contexts, candidates also use AI to generate evaluations of their work, used formatively as an alternative source of feedback beyond traditional supervisor input. Krumsvik (2025) suggests that AI tools can act as 'collaborators' and 'sparring partners'; generating new feedback, enhancing feedback received, and sometimes replacing supervisor feedback due to absence or lack of quality (Krumsvik 2025; Parker et al. 2024). Despite its utility, there is limited research (Bearman et al. 2024b) identifying significant deficiencies in AI tool use, relating to feedback literacy in doctoral environments. We relate this to the concept of feedback literacy, and in our case, feedback literacy in the doctoral environment.

Feedback literacy in undergraduate environments has been reconceptualised in the last five years (Hoo, Deneen, and Boud 2022; Nieminen and Carless 2023; Winstone and Carless 2020; Winstone, Mathlin, and Nash 2019). The reconceptualised concept places students as central and active agents where 'student' enactment prevails over 'teacher' input (Carless and Boud 2018; Dawson et al. 2024; Molloy, Boud, and Henderson 2020). According to Dawson et al. (2024) feedback literacy consists of five domains: students seek feedback information (SF), make sense of information (MS), use feedback information (UF), provide feedback information (PF) and manage affect (MA). In doctoral education, however, there has not been a similar reconceptualisation although there is recent literature exploring this under-researched aspect (Bearman et al. 2024a; Bearman et al. 2024b; Nieminen and Carless 2023). The feedback process in doctoral education remains characterised by traditional methods where supervisors and reviewers provide written comments on thesis components, regardless of discipline (Houston 2018). This feedback is sometimes not understood, misunderstood, vague, unclear, ambiguous and personal (Basic 2021; Carter and Kumar 2017; Hovden and Mjøs 2020; Phan 2024) resulting in high attrition rates, low completions and long candidature periods (Australian Government Department of Education 2019).

As a result of the feedback difficulties students face, AI is now being used to generate formative feedback. Carter and Kumar (2017) suggest access to these capabilities may have a positive impact, particularly among international students, who may be negatively affected by traditional issues of culture-informed power imbalances, a lack of English proficiency, and socio-cultural differences. Students are turning chatbots into podcast hosts, language tutors, assistant professors, and personal trainers. A PhD student in the USA inputted research proposal examiner biographies into a chatbot for oral defence preparation, successfully creating digital simulacrums generating formative feedback as likely viva voce questions (Nature 2025). However, knowing what feedback doctoral students may receive and how it is assessed as constructive and formative remains ambiguous across disciplines and institutions (Grainger, Carey, and Johnston 2024).

A potential solution to formative ambiguity includes providing students with alternative feedback sources, such as formative rubrics aligned to specific research criteria. This is common in undergraduate education but rare in doctoral education (Grainger, Carey, and Johnston 2024). Arguments against rubrics in doctoral education centre on individualistic discipline-specific nature of doctoral research

(Wellington 2013). However, recent literature reports efficacy of formative rubrics in doctoral education, driven by growing acceptance of generic doctoral attributes (Grainger, Carey, and Johnston 2024). Considerable undergraduate research over decades relates to rubric value (Grainger and Weir 2020; Jönsson and Panadero 2017; Reddy and Andrade 2010; Weir 2020) showing how they act as guidelines, eliminate assessor biases, serve as objective quality descriptions, act as discussion starters, interpret standards and illuminate quality characteristics.

Rubric acceptance in doctoral environments relates to assumptions they are only summative scoring tools (Andrade 2023), not aligning with essentially formative doctoral assessment. Hence, there is need to reconceptualise feedback literacy and rethink feedback processes in doctoral education using rubrics as formative tools. These tools are most useful in early doctoral phases where attrition is highest and students are inducted into academia. To emphasise and distinguish the problem of conceiving rubrics as only summative scoring guides, Grainger, Carey, and Johnston (2024) referred to formative rubrics in doctoral contexts as formative assessment criteria-based tools (F.A.C.T).

We investigated generative AI use to provide formative feedback to a native English-speaking doctoral student studying education using the generic F.A.C.T. created by Grainger, Carey, and Johnston (2024) to evaluate a 23,000-word thesis draft in probationary candidature stages. We submitted the draft and rubric into several AI-powered ChatGPT-4 sessions to understand generated feedback, student usage at various stages, feedback actionability, guiding processes, and reliability compared to traditional human reviewer feedback. Although the human reviewer did not use the same rubric as the AI-powered tool, the findings provide insights into our emergent understanding of AI's place in higher education assessment.

Literature review

The recent reconceptualisation of feedback literacy significantly shifts educational research, repositioning students as active agents rather than passive recipients, including student involvement in feedback assessment tool development, design, and use (Jin et al. 2024). Despite research efforts, the doctoral environment remains relatively underexplored, not because doctoral students don't require feedback, but because learning to engage with supervisor and reviewer feedback is crucial to the doctoral process (Bearman et al. 2024b; Carless, Jung, and Li 2024). Research by Bao and Feng (2023) investigated supervisory feedback and doctoral students' academic literacy development, highlighting that current feedback across higher education contexts is predominantly supervisor-driven rather than student-enacted. Both Bao and Feng (2023) and Bearman et al. (2024b) highlight the problematic subjective power-structure doctoral students experience as supervisors can be viewed as brokers, gatekeepers, and judges of their efforts, failings, and successes.

Carless, Jung, and Li (2024) addressed feedback's important role as socialisation in doctoral education emphasising necessary integration of authentic feedback practices. They suggested feedback from sources other than supervisors can prepare students for future academic work. These studies collectively suggest doctoral education has not experienced the same feedback literacy reconceptualisation that

occurred in undergraduate environments. While undergraduate feedback literacy now encompasses five student behaviour domains identified by Dawson et al. (2024), doctoral feedback remains largely characterised by traditional supervisor-driven approaches, contributing to misunderstanding, attrition, power-imbalances, under-developed feedback literacy, and miscommunication among doctoral students (Bao and Feng 2023; Bearman et al. 2024b; Carless, Jung, and Li 2024; Grainger, Carey, and Johnston 2024; Saeed, Al Qunayeer, and Al-Jaberi 2021).

Traditional doctoral feedback processes remain largely characterised by supervisor-enacted talk, text, and formal progress reporting (Bearman et al. 2024b). Bearman et al. (2024b) highlight that feedback within doctoral supervision, while critical to assessment, is often given cursory literature attention, noting the 'intense interpersonal nature of doctoral studies, where relationships between supervisors and candidates span several years' (1). This extended relationship introduces complexities absent in undergraduate settings including lack of formalised criteria in doctoral assessment, resulting in variability in judgments among assessors and reviewers who frequently rely on their own experiences and implicit criteria rather than standardised rubrics found in undergraduate settings (Grainger, Carey, and Johnston 2024). This absence of standardised parameters creates inconsistencies profoundly impacting doctoral candidates' experiences. Parker et al. (2024) referencing Bearman and Luckin (2020) state that traditional assessment methods in doctoral education 'must evolve to account for AI assistance in human compositions' (2), suggesting current supervisor-driven approaches to providing formative feedback undermine educational outcomes among both national and international doctoral students.

Parker et al. (2024) highlighted the potential for AI use in doctoral writing and suggest there is 'a notable lack of research on the specific impact of AI on writing processes and outcomes in doctoral education' (3) suggesting gaps in understanding how these tools might assist doctoral students specifically. Carless, Jung, and Li (2024) uncovered reports of doctoral students using ChatGPT for research design, language proofreading, and quantitative coding, offering indirect evidence of AI's utility for language support. Crompton and Burke (2023) noted that language learning, including writing and reading, are among the most common domains for AI applications in higher education, predominantly at undergraduate levels. These findings collectively support our assertion that AI tools may offer doctoral students valuable additional feedback sources, potentially circumventing implicit barriers related to power imbalances, English proficiency issues, and socio-cultural differences.

Generative AI tools are having significant technological influence in tertiary education with notable variations in adoption between undergraduate and doctoral contexts. AI influence in tertiary education is yielding cautionary responses, particularly around academic integrity issues, with a systematic study by Popenici et al. (2023) reporting distrust from university students and staff in AI, questioning AI's place in educational institutions. Crompton and Burke (2023) systematic review reveals undergraduate students were the most studied demographic in AI higher education research, comprising 72% of focus, with applications predominantly centred on language learning, writing, reading, and vocabulary acquisition. The lack of literature specifically addressing AI impact on writing

processes and outcomes in doctoral education, despite rapid AI tool adoption in academic settings (Parker et al. 2024). This aligns with findings from Grainger, Carey, and Johnston (2024), who document the absence of published research addressing AI in doctoral contexts, particularly around thesis feedback and formative assessment. Rubrics in doctoral education have historically been underutilised despite demonstrated effectiveness in undergraduate contexts. Grainger, Carey, and Johnston (2024) identify a critical gap, pointing out that while doctoral thesis assessment is predominantly formative, few studies reported rubric use for evaluation.

The literature demonstrates progressive feedback approaches at undergraduate levels, contrasting relatively unchanged, supervisor and reviewer-dominated assessment practices in doctoral contexts. The absence of standardised assessment criteria and formalised rubrics creates inconsistencies impacting doctoral candidates, including international students who often face additional linguistic and cultural challenges (Bearman et al. 2024b; Phan 2024). Emerging AI technologies offer promising opportunities beyond traditional barriers related to power imbalances, academic socialisation, English proficiency, and cultural issues. This convergence provides rationale for a single subject case study of a pre-confirmation doctoral student who utilised Grainger's formative assessment criteria-based tool (F.A.C.T.) with ChatGPT to assess and obtain feedback before sending work to confirmation reviewers.

Our specific research questions were:

1. How can AI tools be used to enhance feedback practices in the probationary doctoral context?
2. How accurate is Grainger's Formative Assessment Criteria-Based Tool (F.A.C.T.) when combined with AI?
3. Is AI a reliable formative feedback mechanism for doctoral students?
4. How does AI generated feedback align with the feedback given by reviewers?

Methods

The single-subject case study adopts a naturalistic and interpretive approach made popular by Robert Stake (1995). This methodology prioritises holistic understanding through immersion in natural context and allows the researcher's interpretive insights to highlight case characteristics. We provide an overview of how Grainger's Formative Assessment Criteria-Based Tool (F.A.C.T.) was used with ChatGPT to assess emerging versions of a pre-confirmation thesis document for feedback. In Australia, the pre-confirmation stage can be likened to a probationary stage when candidates write research proposals for official endorsement by reviewers and supervisory teams.

The paid ChatGPT Plus version of GPT-4 was used as it allowed extended enquiries and creation of custom GPTs. Data gathered includes downloaded offline history of all ChatGPT interactions dated prior to 2024. While maintaining the naturalistic and interpretive foundation, this study also applies a sequential design approach complementing Stake's (1995) methodology by providing structured yet flexible framework. This combination allows emergent understanding through progressive

phases, with each phase building upon findings from the previous, enabling both holistic interpretation and systematic assessment of ChatGPT's capabilities within the case study framework (Chelli et al. 2024).

The initial test was conducted in May 2024 with creation of a private GPT chatbot labelled *Thesis Helper*. This private chatbot held a current 23,000-word draft of the case study pre-confirmation thesis (including references) and allowed questions without uploading the thesis for every chat. The document submitted was written as three chapters: Introduction, Literature Review, and Methodology. We started a new private *Thesis Helper* chat and uploaded the F.A.C.T. rubric in a word document (refer Appendix C). We wanted to ensure ChatGPT could read the docx file and understand what it was examining. Large Language Models (LLMs) like ChatGPT have capacity to hallucinate and make mistakes (Chelli et al. 2024). A documented way of testing and verifying an LLM's ability to understand input is through human user verification of the LLM's understanding (Song et al. 2025). The chatbot replied with comprehensive and accurate rubric summary, demonstrating its ability to read all table headers and content (refer Appendix A).

Following the verification of understanding, the chatbot was asked to provide feedback on the embedded thesis based upon the rubric. These chatbot questions, also known as prompts, were suggested by the author. Questions were applied including requests that asked ChatGPT to take on a single persona – an approach used by Codone (2024) who directed ChatGPT to take on the persona of a school superintendent when assessing student work. In our study, ChatGPT provided structured feedback based upon rubric components and criteria identified. It commenced by repeating the thesis title and proceeded to provide feedback for each F.A.C.T. rubric criteria C1 through C7: C1 Significance of research, C2 Knowledge of sources, C3 Validity of Methodology, C4 Rationale/justification for the research. C5 Contribution to knowledge. C6 Writing according to research paradigm and discipline, C7 Progress as a researcher. Each feedback portion provided by ChatGPT was between 40 and 55 words in length. The feedback is discussed in the results section (refer Appendix B).

This initial test response was encouraging; however, ChatGPT did not provide clear idea of what standard each criterion was mapped against. According to Grainger's F.A.C.T. rubric, three standards were possible: *meets expectations, needs improvement*, or *unsatisfactory*. As a subsequent step, we created our own private GPT chatbot titled *Formative Rubric*. Instead of uploading the rubric to a thesis-based GPT chatbot, we wanted to explore what might happen if we did the opposite and uploaded a thesis to a rubric-based GPT chatbot. The *Formative Rubric* chatbot was created to provide constructive feedback of uploaded thesis documents based on the F.A.C.T. rubric criteria. The feedback was to be presented in formal academic language and follow APA referencing guidelines. This approach is similar to one taken by Krumsvik (2025) who created a chatbot named *Writing the Synopsis Companion for PhD-candidates*.

Several 'verification understanding' prompts were conducted to check the chatbot for accuracy, and upon assessing them as correct, several chat sessions were conducted across May 2024 requesting feedback from the *Formative Rubric* AI chatbot. The chatbot feed from the pre-confirmation doctoral thesis provided varied results

depending on the prompts and therefore allowed us to compare the feedback gathered from the Independent Academic Reviewer (human) who was sent the case study thesis for review, according to the confirmation review guidelines provided by the institution.

The thesis submissions sent to reviewer was not mapped by an Independent Academic Reviewer (human) against Grainger's F.A.C.T. rubric. As there are few formative assessment rubrics across Australian universities for doctoral students, institutions commonly create their own assessment methods and materials (Grainger, Carey, and Johnston 2024). In this case study, the institution provides a Microsoft Word template to reviewers containing ten sections for reviewers' feedback: General comments, Originality, Feasibility, Theoretical foundation, Literature review, Methods, Critical thought, Contribution to knowledge, Edits and referencing, and Overall recommendation. In this case study, an external reviewer provided feedback. The process is represented in Figure 1 which illustrates the sequential design process commencing with the creation of the *Formative Rubric* chatbot in ChatGPT, the verification process, and the submission of the pre-confirmation thesis document to both the chatbot and Independent Academic Reviewers.

Results

The feedback gathered from both the *Formative Rubric* AI chatbot and the Independent Academic Reviewer based on the relative prompting and submission of the pre-confirmation thesis reveals noteworthy patterns and distinctions. This section presents findings through an interpretive lens, allowing us to examine feedback provided by AI and humans. Stake's (1995) interpretive approach was selected over alternate methodologies such as thematic analysis or sentiment analysis, as it enables us to contextualise understanding of different feedback sources from both AI and human reviewers. This methodological choice aligns with our research objectives of not only comparing feedback content but considering practical implications, specifically, the actionable steps a doctoral student might take based on each feedback source.

Results: AI chatbot

The first attempt to gather feedback from a pre-confirmation thesis draft was within the context of a custom-made private ChatGPT chatbot named *Thesis Helper*. After

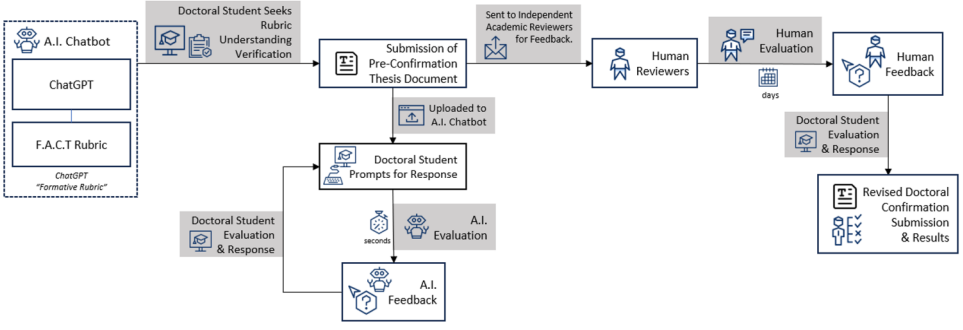

Figure 1. Sequential process used in case study (Source: authors).

confirming the chatbot could read the document, the F.A.C.T. rubric was uploaded. The results in Appendix A confirm that ChatGPT could read and respond in a way that confirmed the contents of the rubric. Adopting a persona, the student then asked the chatbot to provide feedback. The chatbot went beyond simply explaining that it could read the rubric document but could use it to assess the thesis by providing a sentence or two for each of the seven rubric criterion. In its final remarks, the chatbot said:

> Overall, your thesis meets the high standards expected for a PhD dissertation in terms of depth, analytical rigor, and contribution to the field. It demonstrates a robust understanding of both the theoretical framework and the practical implications of your research, aligning well with the specified criteria in the rubric.

Using the *Formation Rubric* chatbot, chat sessions were tested with varied prompts and responses. These are listed in Table 1 displaying the chat session number, student chat input, and summary of AI chatbot results. There were multiple chat sessions for two main reasons. First, chats were commenced on different days. Second, new chat sessions allowed for testing of chat actions without drawing on previous results from the same chat thread, allowing the student to evaluate 'hallucinations' and misinformation in areas of suggestions (Chelli et al. 2024). Student chat actions were chosen to mirror similar testing of personas (Codone 2024; Phan 2024), testing for accuracy (Krumsvik 2025), and request feedback (Hirunyasiri et al. 2023).

Results: feedback from reviewers

A reviewer feedback template was provided by the PhD Student host-university for the reviewer to use. However, this template contained no section for providing overall standard of *meets expectations, needs improvement,* or *unsatisfactory.* Instead, a second feedback form allowed the reviewer to highlight the standard. The reviewer awarded the student *needs improvement.* The reviewer provided feedback of approximately 2300 words which were not evenly weighed across sections with no clearly defined action steps or references provided. The initial General Comments section contained over 1500 words, making up 65% of feedback. This reviewer awarded a *Needs Improvement* standard, concluding:

> Needs improvement. I would encourage the candidate to consider the theoretical contribution and methods. Such a process will also necessitate a reshaped Literature Review. Overall, I would like to reiterate that I can see the level of work in which the candidate has engaged. I just think that the current juncture is an appropriate time to raise such issues, while it is still possible to establish a sound research trajectory for the work.

In summary, the reviewer suggested several improvements: developing robust theoretical foundation before data collection, refining research question and strengthening methodology through deeper qualitative approaches, positioning study at intersection of alternative approaches, demonstrating analytical depth where possible at current research stage, and acknowledging the document was well-structured and appropriately referenced.

Table 1. ChatGPT sessions.

Chat session number	Student chat action	Summary of ChatGPT result
1	'Tell me how this rubric works'	ChatGPT provided a summary of the seven criterion and three levels of achievement as listed in the rubric. It also provided a summary of the role of the rubric as a standard assessment tool.
2	'Imagine you are a thesis reviewer in academia. If I upload a document, can you provide feedback based on the formative rubric?'	ChatGPT prompted for the file to be uploaded. Once uploaded it provided feedback on the document attributing *Meets Expectation* across all 7 criteria.
	'Imagine you are a supervisor using the rubric as a guide, what advice would you give me for improvement?'	ChatGPT provided 1–2 sentences per criteria. The language was constructive. Some examples include: C2: 'Your literature review is comprehensive but could benefit from a broader scope that includes global perspectives or comparative studies, particularly from [details omitted]. This could help position your research within a global context and highlight unique contributions or deviations in the Australian setting'. C5: 'Deepen the analysis in your discussion section by comparing your findings with existing theories and research. Highlight contradictions or confirmations with established knowledge and discuss the theoretical and practical implications in greater detail'. It concluded: "Implementing these suggestions should help refine your thesis, ensuring that it not only meets but exceeds the expectations of the confirmation rubric, thereby making a significant scholarly contribution to the field of business management and education."
	'Are there other studies, you know of, similar to what I am proposing?'	ChatGPT suggested that the research was positioned within an emergent but growing field and went on to cover seven similar research initiatives and frameworks including four journal article references by name and year.
3	'Can you use the rubric to give me marks in each criterion? Give me a prompt to upload my file, then list results in a table and simply give me the standard mark. For example, Meet expectations, need improvement, unsatisfactory. Also create a row with feedback on why I received that mark'.	ChatGPT replied with a table as requested. In this case five of the seven criteria received a *Meets Expectation,* and two *Needs Improvement*. Just a single line was provided in the Feedback column. E.g. C5: Contribution to knowledge | *Needs Improvement* | The contribution to knowledge is stated but could be enhanced by more explicitly defining and evidencing the originality and novelty of the study's insights.
	"This doesn't seem critical enough. Can you re-assess it?"	ChatGPT, reading the same thesis, in the same session awarded two *Meets Expectation,* four *Needs Improvement,* and one *Unsatisfactory.* The feedback included more critical comments. E.g. C5: Contribution to knowledge | *Unsatisfactory* | The thesis proposes contributions to knowledge, but these claims are not adequately supported by evidence of originality or a clear demonstration of how they advance the field. To meet expectations, you should clearly articulate the novel aspects of your research and support these claims with more concrete examples or preliminary data.

(*Continued*)

Table 1. Continued.

Chat session number	Student chat action	Summary of ChatGPT result
4	'Can you use the rubric to give me critical marks in each criterion? Give me a prompt to upload my file, then list results in a table and give me the appropriate mark. For example Meet expectations, needs improvement, unsatisfactory. Also create a row with feedback on why I received that mark. I want you to be encouraging but still objectively critical with your marking and feedback, like a professor would who is considering the research as a worthy pursuit. List the feedback in dot points'.	Upon uploading the document ChatGPT created a table similar to the previous sessions but added recommendations to the Feedback column. E.g. C5: Contribution to Knowledge | *Needs Improvement* | – States the contribution to knowledge, but evidence and originality of contribution could be detailed more explicitly. – Recommend enhancing the discussion on how this research advances new insights or theories. Below the table, ChatGPT provided "Encouragement and Additional Feedback:" including Strengths, Areas for Improvement, and General Note.

Discussion

When considering AI as an enhancement to traditional feedback practices we note from the AI chatbot results that ChatGPT sessions demonstrated potential to provide supplementary formative feedback with exceptional efficiency. The paid ChatGPT Plus version allowed uploading of large documents for analysis, including a 23,000-word thesis and rubric, allowing the student to query both. When considering feedback's important role in doctoral education, we suggest, based upon differing opinions of AI personas and human reviewers, that AI could play an important role in doctoral student socialisation which Carless, Jung, and Li (2024) address. Perhaps it is not a question of 'either/or', but a supplementary 'both and more' when it comes to AI enhancement. Furthermore, the control which the student has over the AI chatbot's critical nature *via* persona requests also demonstrated a way in which problematic subjective power-structures may be eliminated (Bearman et al. 2024b; Parker et al. 2024).

When combined with AI, the rubric used in this study, Grainger's Formative Assessment Criteria-Based Tool (F.A.C.T.), yielded accurate results on recall demonstrating ability to read the rubric and apply it to the thesis (See Appendix A and Table 1). Cases of doctoral students creating their own AI formative feedback tools that emulate reviewers have been recently documented (Nature 2025), highlighting not only desire for AI-driven feedback but also doctoral students' ability to leverage LLMs like ChatGPT. The results and application of a rubric in doctoral studies may offer a partial solution to the 'absence of standardised assessment parameters' in doctoral education which Grainger, Carey, and Johnston (2024) identified as a critical gap. However, justifying the use of an AI chatbot and rubric, which has not been adopted by a higher education institution, simply because 'one can' may, in fact, be counterproductive and harmful to the student's progress, socialisation, and feedback experience. If doctoral socialisation is a human experience (Carless, Jung, and Li 2024), non-human feedback provided by AI to human doctoral students may not prepare students for the as co-constructed, relational, and socio-cultural elements

of academic life beyond rubrics and confirmation documents. That said, without documenting the student experience, this question of student preparation remains inconclusive.

Next, we consider if AI is a reliable formative feedback mechanism for doctoral students. If reliability were to be measured as being consistent answers across times and chat sessions, the varied results of feedback in Chat Session number 2, 3, and 4 in Table 1 indicate that AI is not a reliable mechanism. There was clear inconsistency in ways the chatbot awarded *Meets Expectation*, *Needs Improvement*, and *Unsatisfactory* across rubric criteria between chat sessions. This unreliability confirms findings in existing literature about poor feedback reliability in doctoral contexts (Parker et al. 2024) and AI's propensity towards 'hallucinations' and misinformation (Chelli et al. 2024). On the other hand, if reliability were considered in terms of expediency and timely completion, this study would support Zapata et al. (2024) whose comparative study between AI and human reviewers yielded similar results around unmatchable speed that AI demonstrated. Research by Neupane Bastola (2022), highlights sentiment among doctoral students who 'have to waste their time waiting' (9) for feedback from academic supervisors and reviewers. Recent studies among undergraduate students also highlight the negative impact that extended waiting periods can have on student motivation (Fisher et al. 2025). Fisher et al. (2025) posit that, 'The challenge for educators and institutions is to find a balance where feedback is delivered promptly enough to be relevant yet is also detailed and personalised enough to be truly beneficial. In practice, achieving this balance might involve adopting a tiered approach to feedback where initial, quick responses provide immediate guidance, and more detailed feedback follows once educators have had the time to thoroughly assess the students'. (8). There may be a place for AI generated feedback in providing less nuanced and contextual responses quickly alongside human feedback. This may reduce the sense of wasting time and sustain motivation among doctoral students, particularly if there is some continuity in the ways which AI and human reviewers assess doctoral work, such as a formative rubric.

Deidentification and privacy concerns limit what can be documented in this article, however when asking how AI generated feedback aligns with feedback given by reviewers, our interpretive examination of feedback from both AI and the human reviewer revealed notable patterns - both convergence and divergence, reflecting variability in judgments among assessors that Grainger, Carey, and Johnston (2024) identified as a persistent challenge in doctoral assessment. When comparing AI-generated feedback with responses provided by the human reviewer in this study, several significant interpretations emerge. The AI consistently identified similar areas needing improvement as human reviewers, particularly regarding theoretical foundation and contribution to knowledge. The Reviewer and the AI (especially in Chat Session 3, Table 1) highlighted concerns about clarity of the research's contribution to theory, with the AI noting the thesis 'proposes contributions to knowledge, but these claims are not adequately supported by evidence of originality'. This mirrors the Reviewer comment that they experienced difficulty in identifying a clear knowledge contribution. However, significant differences were apparent. The human reviewer provided more contextually nuanced feedback, with specific disciplinary insights that the AI did not. Additionally, human reviewers offered more actionable

and specific suggestions for improvement. AI's suggestions, while thematically similar, remained more general and less tailored to disciplinary practices.

Conclusion

This study examined how a single probationary doctoral candidate utilised AI-enhanced mechanisms to solicit feedback using Grainger's Formative Assessment Criteria-Based Tool (F.A.C.T.) as a rubric. AI-generated feedback demonstrated some alignment with human reviewers in identifying areas needing improvement, particularly regarding theoretical foundation. However, the human reviewer provided more contextually discipline-specific and nuanced feedback with more actionable suggestions. The study illustrates how generative AI, when integrated with formative rubrics, may serve as a supplementary feedback mechanism providing fast formative feedback with potential to increase feedback literacy, prepare doctoral students for human feedback, and reduce power imbalances often reported in traditional supervisor/reviewer-student relationships. While this study provides valuable insights, its findings are limited as they are derived from a single-subject case study, making it difficult to apply conclusions broadly to other populations or contexts. This study only tracked activities of a single Australian-educated and native English-speaking individual, with a single thesis of certain style and language, within a single regional Australian university, in a single discipline, with feedback from a single human reviewer, over a small time-period. These limitations made it impossible to consider implications from engaging with generative AI prior to human reviewers as an international student, and/or someone from a non-English-speaking background, who may experience feedback differently. Further research may yield results in how AI tools can act as 'collaborators' and 'sparring partners' (Krumsvik 2025) which may improve English proficiency and assist in navigating power-imbalances by preparing a wider range of students for critical feedback by preparing them for what may result from submission to the reviewers (Bearman et al. 2024b; Carter and Kumar 2017; Parker et al. 2024). The scope of this study also presented limitations around further uses for AI in formative assessment, both present and emerging.

Many gaps remain across literature in both formative assessment of doctoral students and the legitimacy of AI in higher education. In this study, we were not able to compare AI's use of the F.A.C.T rubric alongside human reviewers using the same rubric. Higher education institutions such as universities willing to integrate formative rubrics into their doctoral confirmational processes would provide further comparable insights, however this study was performed in an assessment setting most common among universities; that being, doctoral assessments absent of standardised rubrics. Variations to further research using formative rubrics such as F.A.C.T., AI, doctoral students, and human reviewers may include reflective thoughts and evaluation from reviewers and doctoral students on F.A.C.T. use as a formative tool (Grainger, Carey, and Johnston 2024).

Although warranted in a single-subject case study, the student experience has not been documented here. Further research may explore variance and outcomes of self-authored AI prompts created by doctoral students, alongside feedback on whether students found the AI feedback supportive to their learning. Also, submission of

more than one thesis to an AI chatbot like the *Formative Rubric* chatbot created for this study to assess its reliability and nature of feedback may yield valuable results alongside the prompt variations used in the AI chatbot interface. Therefore, we conclude that it is possible for reflective practice in doctoral education to be enhanced through AI integration, addressing gaps in feedback literacy, socialisation, and standardised assessment parameters in doctoral contexts, however further research is required in this emerging field.

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Appendices

Appendix A – Verification of understanding

In order to verify ChatGPT's understanding of the F.A.C.T. rubric we asked it the following question:

User: Can you explain what you see on the table?

ChatGPT: The table in your rubric document presents a structured evaluation criteria for PhD theses, aligned with seven major components, categorised into three performance levels: 'Meets Expectations', 'Needs Improvement' and Unsatisfactory'. Here's a summary of what each category assesses:

1. **Significance of Research (C1)**: Evaluates the clarity in situating the study within a broader societal or academic context, and the effectiveness in demonstrating how the research addresses critical issues.
2. **Knowledge of Sources (C2)**: Assesses the range, relevancy, and critical evaluation of cited sources, and their alignment with the research questions.
3. **Validity of Methodology (C3)**: Considers the clarity, justification, and appropriateness of the methodology in relation to the research objectives and questions.
4. **Rationale/Justification for the Research (C4)**: Reviews the rationale provided for the research through critical literature review and the demonstration of the research's potential contribution.
5. **Contribution to Knowledge (C5)**: Examines the clarity and justification of the study's contribution to new knowledge or insights.
6. **Writing According to Research Paradigm and Discipline (C6)**: Measures the coherence, accuracy of terminology, and logical structure of the narrative within the context of the specific research paradigm and discipline.
7. **Progress as a Researcher (C7)**: Focuses on the researcher's independence, response to feedback, understanding of ethical considerations, and overall progress in their research journey.

Appendix B – *Initial criterion feedback*

C1 – Your thesis situates the study within the significant context of [the topic]. This effectively demonstrates the research's importance and aligns well with the rubric's criteria for meeting expectations.

C2 – The thesis presents a well-researched foundation, citing a wide range of relevant sources that critically evaluate and integrate into your research framework. Your literature review discusses the gaps within current studies… supporting the thesis's arguments with up-to-date and seminal sources.

C3 – You have clearly explained and justified your mixed-methods exploratory case study approach. The use of qualitative and quantitative methods, including interviews and surveys, is well-suited to your research questions, aiming to capture both tangible and intangible benefits of [the topic]

C4 – The thesis offers a compelling rationale for the research by highlighting the lack of focused studies on [the topic]. This is effectively justified through a critical review of relevant literature and identification of a significant research gap.

C5 – Your work defines and justifies how it advances knowledge by [the topic].

C6 – The thesis is coherent, logically structured, and well-written, adhering to the conventions of qualitative research and the discipline's methodological frameworks. The use of discipline-specific terminology is accurate and enhances the readability and academic integrity of your work.

C7 – The independence you exhibit in identifying a research gap, formulating research questions, and choosing appropriate methodologies shows significant progress as a researcher. Your approach to integrating feedback and ethical considerations into your research process aligns well with the expectations outlined in the rubric.

Appendix C – *Grainger's formative rubric*

UniSC Standard	C1 Significance of research	C2 Knowledge of sources	C3 Validity of Methodology	C4 Rationale/justification for the research	C5 Contribution to knowledge	C6 Writing according to research paradigm and discipline	C7 Progress as a researcher
Meets Expectations	Situates the study, clearly identifying the importance of the research within a broader societal or academic context; effectively demonstrates the research addresses critical issues.	Cites a wide variety and a wide range of relevant sources, seminal and up to date from policy and non-policy literature that clearly align to the research questions or topic; critically evaluates sources and effectively integrates literature into the construction of the research framework.	Explains clearly and justifies appropriately and in detail, a well-chosen, novel and rigorously designed methodology that aligns with the topic/ research objectives/aims/research questions to ensure validity and reliability of findings.	Articulates a clear, compelling rationale/justification for the research by critically reviewing relevant literature and clearly demonstrating the research's potential to contribute significant insights or practical solutions to the field. Presents convincing arguments about importance, making a strong case for its necessity and potential impact.	Clearly defines and justifies explicitly, using a range of reasons, why and how the study will advance/contribute to new knowledge/insights/theories ; provides a range of evidence of an original/novel contribution that adds to the existing academic or practical knowledge base.	Writes and presents information coherently, consistently aligning with the conventions and methodologies of their research paradigm and discipline; uses discipline-appropriate terminology and concepts accurately; structures the narrative logically, showing effective coherence and cohesion, which enhances the readability and academic integrity of the work.	Evidences 'independence' and demonstrates progress as a researcher and is on-track to meet future milestones (e.g., defending choices, meeting all deadlines ahead of time, responding to feedback and rectifying issues, understanding ethical considerations, data collection, completing relevant training courses, completing a viable research plan with appropriate budget).
Needs Improvement	Situates the study mentioning the significance of the research but does not effectively connect it to broader issues.	Cites key/ relevant/ seminal sources, but application to the research is often tangential.	Describes an appropriate methodology that aligns with isolated aspects of the topic/research objectives/aims/research questions.	Provides a clear, valid, robust and sometimes convincing rationale/justification for the research, hinting at necessity, relevance, potential impact, significance, importance and anticipated contributions to the field, albeit sometimes vague but aligned with disciplinary expectations.	States valid reasons regarding why the study will contribute to knowledge; provides relevant evidence of a valid and original contribution to knowledge.	Writes inconsistently and with occasional minor errors but shows a fundamental understanding of the research paradigm and the key disciplinary conventions; underuses or incorrectly applies discipline-specific concepts, disrupting coherence.	Evidences isolated aspects of 'independence' as a researcher, seeks advice, meets key deadlines, acknowledges key aspects of feedback.
Unsatisfactory	Situates fragmented aspects of the study, identifies isolated and/or insignificant or unsubstantiated claims regarding why the research needs to be done.	Cites and evaluates isolated and/or dated literature, indicating insufficient knowledge of relevant sources.	Describes inappropriate methodology, explains isolated and/or fragmented aspects of the methodology (e.g., states inappropriate and/or invalid research questions that are misaligned with the rationale/significance or purpose of the study.	The rationale for the research is unclear, vague and/or poorly developed, with minimal alignment with the study aims.	Gives invalid, unsubstantiated or unrelated reasons and claims regarding why the study will contribute to knowledge, or how the study is original and significant.	Writes inappropriately for the genre, using minimal or incorrect discipline-specific terminology, resulting in disjointed narrative that disrupts cohesion; ideas are often disconnected.	Relies on supervisors; needs remedial action/strategies to ensure future milestones can be met, reacts to isolated aspects of feedback.